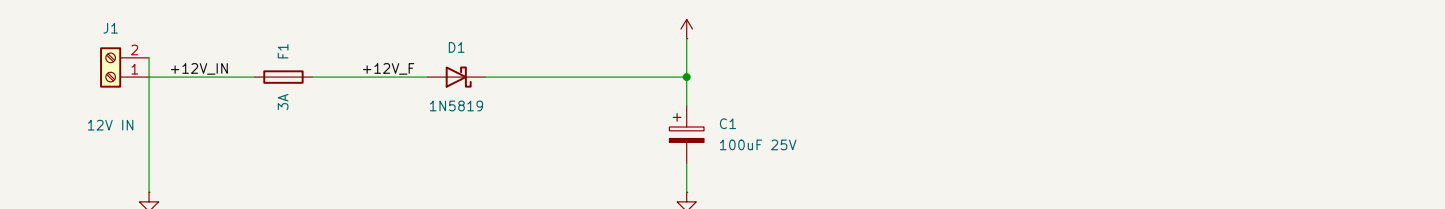
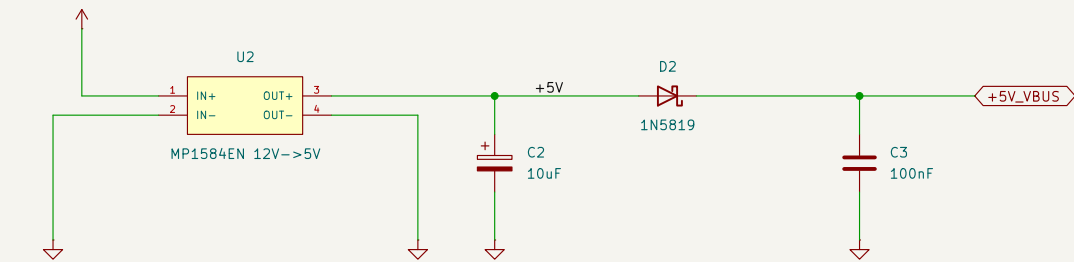


1. 12 V INPUT AND PROTECTION



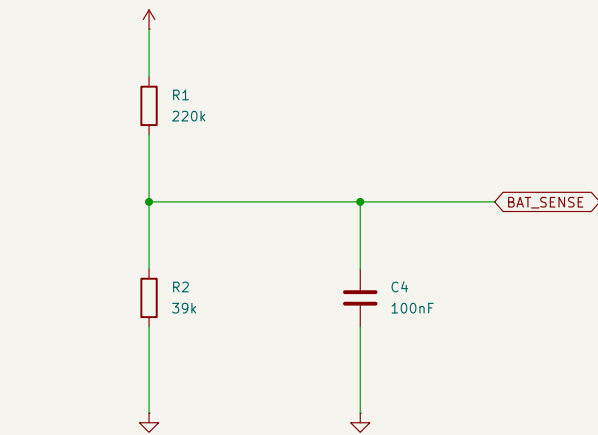
D1 is series reverse-polarity protection, costing ~0.4 V – a 12 V solenoid still operates at 11.6 V. F1 protects the BATTERY, not the board: a shorted valve cable across a lead-acid bank is a fire risk and nothing else here limits that current.

2. 5 V SUPPLY



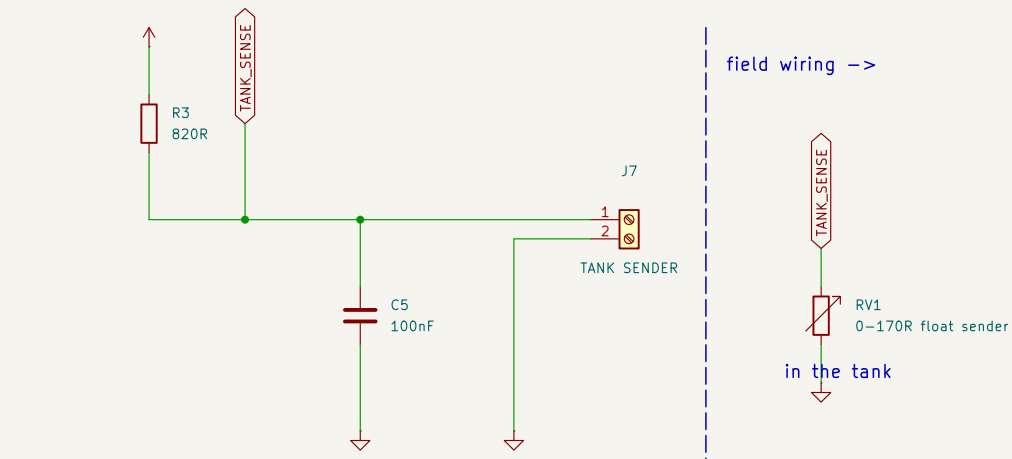
D2 IS NOT OPTIONAL, as the buck setting is part of it. The S2 Mini ties VBUS straight to USB 5 V, so both supplies meet on one node the moment someone plugs in to flash the board. Set the buck to 5.2 V: after D2 the rail is -4.9 V, just BELOW USB, so when USB is present it wins and nothing is driven back into the host port. D2 also stops USB back-feeding the buck. Do not set the buck above -5.3 V or it will push current into the host through D2.

3. BATTERY SENSE -> GPI08



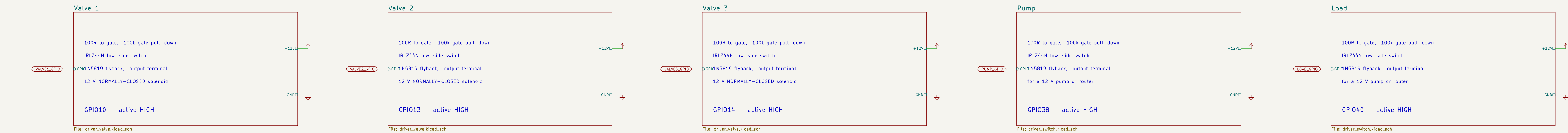
Ratio $(220+39)/39 = 6.1$ – SENSOR_BATTERY_VOLTAGE_DIVIDER must match, and it takes a float
 15 V gives 2.26 V at the pin, inside the S2 ADC's 0.2500 mV linear range; 12.6 V gives 1.90 V.
 High values because a 12 V bank moves slowly, so 49 uA of standing drain is worth more than
 resolution. C4 is required, not optional: source impedance is $220k||39k = 33 k$, far above the
 ADC's 10 k preference, and $\tau = 3.3$ ms against a 10 s sample interval.
 With the lower leg fitted an absent 12 V rail reads ~0 mV rather than floating, which is why
 SYSTEM_POWER_LED_READING_MV=9000 is safe and the old "comment it out on USB" note is gone.

4. TANK LEVEL -> GPIO6



R3 and the sender divide 3V3: 0R empty, 170R full -> 0.567 V full. An OPEN circuit pulls the pin to 3V3, which the firmware reports as "no sender fitted" rather than as an empty tank - the distinction ODSPI could not make, and why a missing sender does not block irrigation. The resistances are nominal and were never measured, so calibrate: Tare with the tank empty, then enter 100 with it full. Run the sender on shielded or twisted cable.

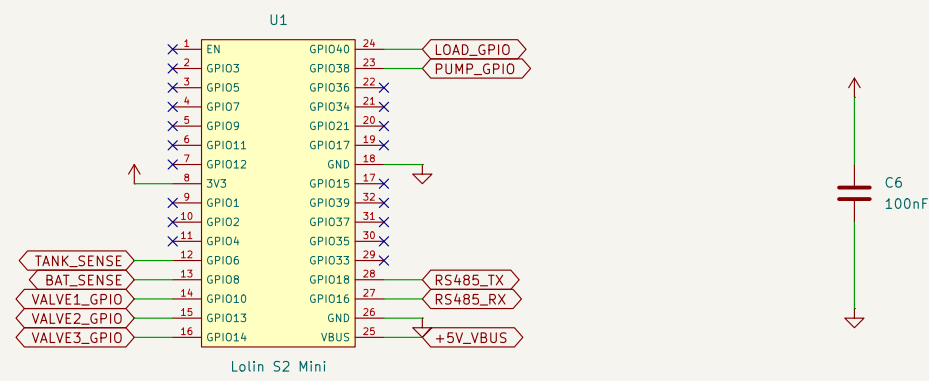
8. OUTPUT DRIVERS – one block per channel. Only GPIO, +12 V and GND leave a block; the driver is on pages 2–6



Valves **MUST** be **NORMALLY CLOSED** and must not be latching. Loss of 12 V then closes them mechanically, and that – not the firmware – is what guarantees a dead controller cannot leave water running. A latching valve holds its last state and the firmware cannot drive one anyway.

PUMP and LOAD switch the low side of a 12 V device. For a MAINS pump or router, drive an external Sonoff from the frugal-iot server instead: nothing on this sheet is rated for mains, and mixing mains with low voltage raises the safety bar a long way beyond what this drawing covers.

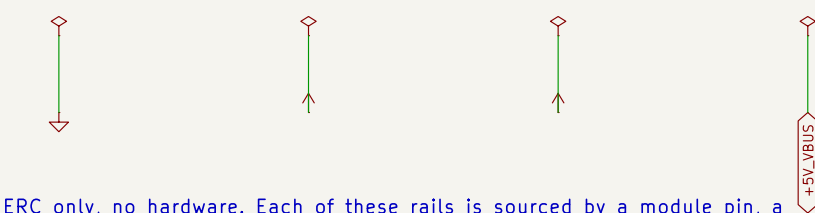
6. CONTROLLER



Thirty-two pins on one part: drawn with net labels, which is how an MCU is normally shown. The blocks that are real circuits – the input, the supply, the two dividers and each output channel – are drawn as circuits.

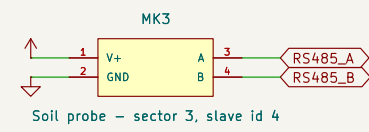
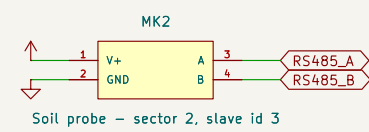
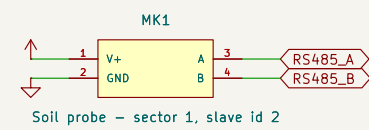
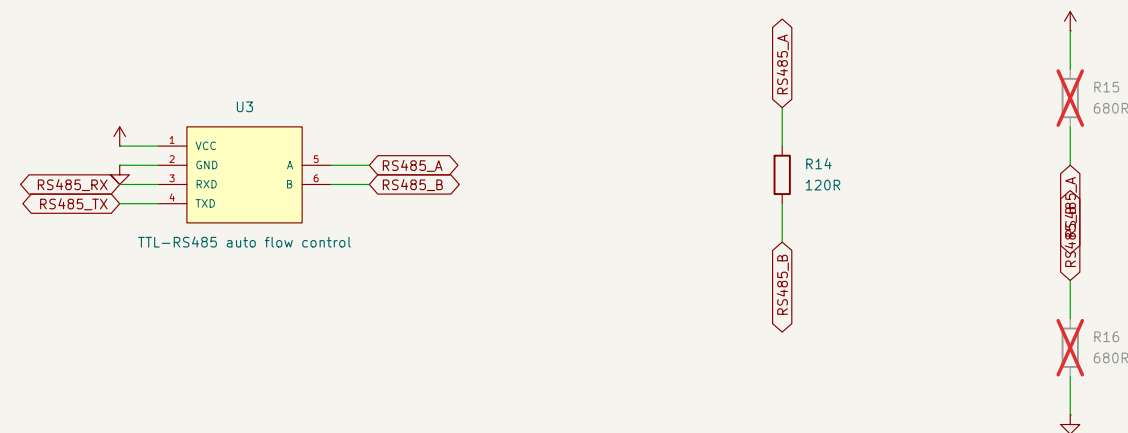
Powered through VBUS (pin 25), never through the 3V3 pin: feeding 3V3 directly would back-drive the module's own regulator. 1.2 V to 3.3 V with an LDO is not an option either – at the S2's ~300 mA WiFi peak that is $(1.2-3.3) \times 0.3 = 2.6$ W, which needs a heatsink. Unused GPIOs are marked no-connect. GPIO0/45/46 are the S2 strapping pins and none of the pins used here are among them, so nothing here affects boot mode.

5. POWER FLAGS



ERC only, no hardware. Each of these rails is sourced by a module pin, a connector or a power symbol rather than by a regulator on this sheet, so ERC needs telling that they really are driven.

7. RS485 BUS AND SOIL PROBES



THE MODULE MUST SWITCH DIRECTION ITSELF. The firmware defines no DE/RE pin, so `System_RS485::txEnable()` does nothing at all – a plain MAX485 breakout with a DE/RE jumper will receive but never transmit. Search for the words "automatic flow control".

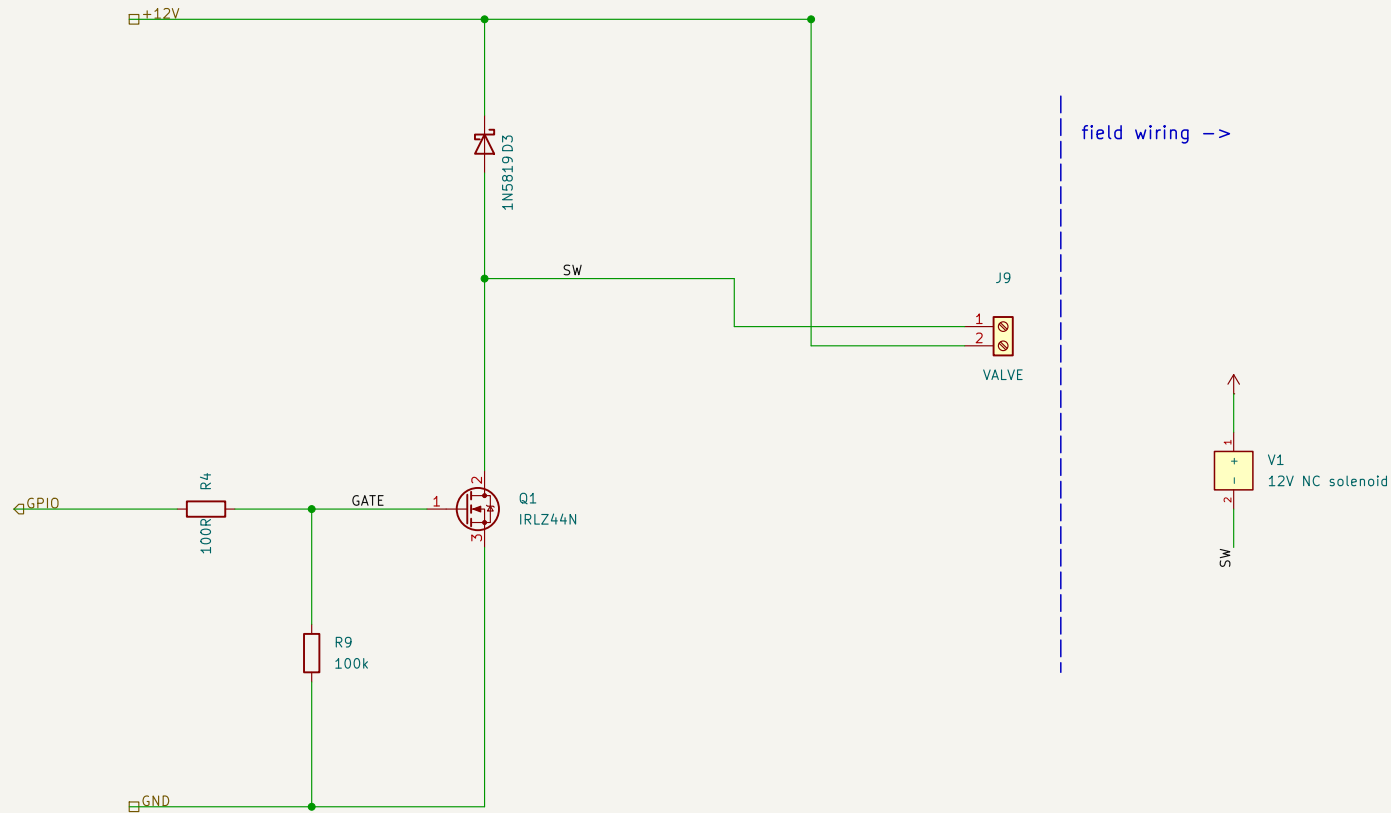
R14 (120R) belongs at the FAR END of the bus, at the last probe – not on this board. Fit ONE termination only. With 120R at both ends the pair below gives just $3.3 \times 60/1420 = 139$ mV of idle differential, under the 200 mV the standard asks for; use 470R instead.

R15/R16 are fail-safe bias and are marked DNP because most auto-direction modules already fit bias and termination. CHECK YOURS before adding a second set.

Probes accept 5–30 V so they run straight off the 12 V rail. Slave ids are auto-provisioned from 2 upwards, so connect the probes ONE AT A TIME, in sector order – address 1 has to keep meaning “not yet provisioned”. Field cable: shielded twisted pair, one pair A/B and one pair 12 V/GND, shield earthed at the controller end only.

ONE VALVE CHANNEL

Drawn once here and used three times – see the root sheet.



GPIO, +12 V and GND are the only connections that leave this sheet. Everything else is internal, which is why the root sheet can draw the whole channel as one block.

RPD (100k, gate to GND) is a SAFETY part, not decoration: ESP32 GPIOs float before setup() runs and through a reset or a brownout, and this resistor is what holds the output off across that window.

Active HIGH is fixed by the firmware – Actuator_Digital::act() is a bare digitalWrite(pin, value) with no invert flag. A low-side N-MOSFET is the natural match; most cheap opto-relay boards are active LOW and would invert every output.

D clamps the inductive kick when the FET turns off. Cathode to +12 V – backwards, it is a short across the supply the moment the output goes on.

Sheet: /Valve 1/
File: driver_valve.kicad_sch

Title:

Size: A4 Date:

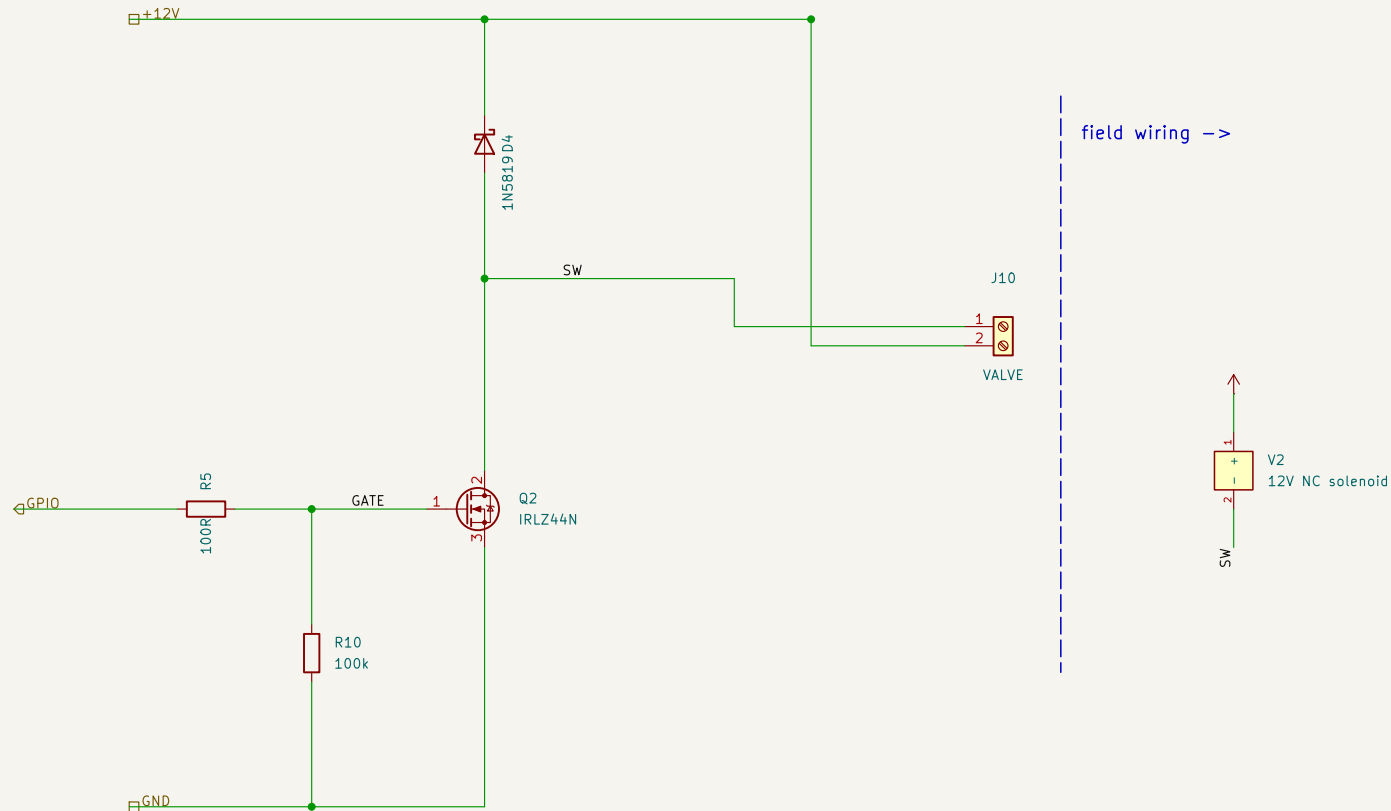
KiCad E.D.A. 10.0.4

Rev:

Id: 2/6

ONE VALVE CHANNEL

Drawn once here and used three times – see the root sheet.



GPIO, +12 V and GND are the only connections that leave this sheet. Everything else is internal, which is why the root sheet can draw the whole channel as one block.

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Sheet: /Valve 2/
File: driver_valve.kicad_sch

Title:

Size: A4 Date:

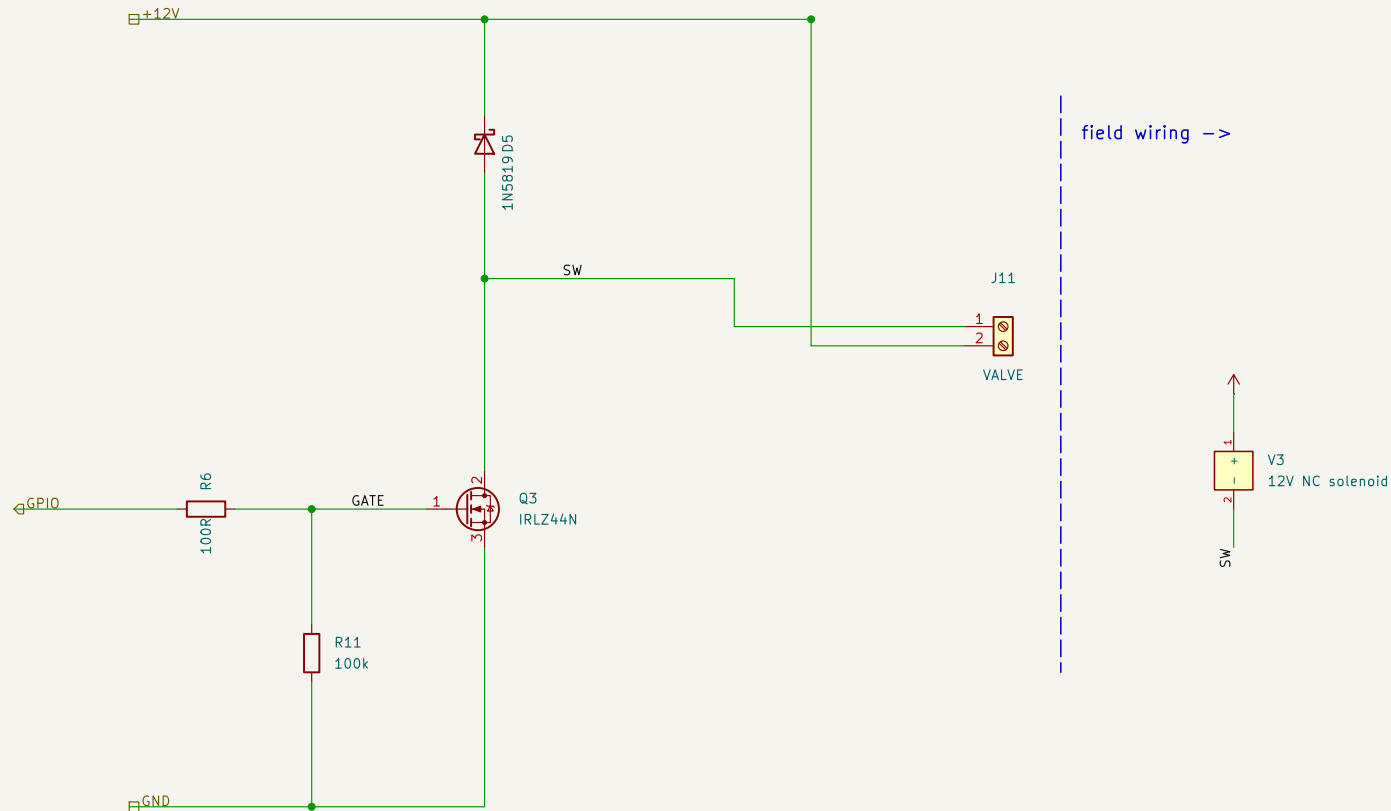
KiCad E.D.A. 10.0.4

Rev:

Id: 3/6

ONE VALVE CHANNEL

Drawn once here and used three times – see the root sheet.



GPIO, +12 V and GND are the only connections that leave this sheet. Everything else is internal, which is why the root sheet can draw the whole channel as one block.

RPD (100k, gate to GND) is a SAFETY part, not decoration: ESP32 GPIOs float before setup() runs and through a reset or a brownout, and this resistor is what holds the output off across that window.

Active HIGH is fixed by the firmware – Actuator_Digital::act() is a bare digitalWrite(pin, value) with no invert flag. A low-side N-MOSFET is the natural match; most cheap opto-relay boards are active LOW and would invert every output.

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Sheet: /Valve 3/
File: driver_valve.kicad_sch

Title:

Size: A4
KiCad E.D.A. 10.0.4

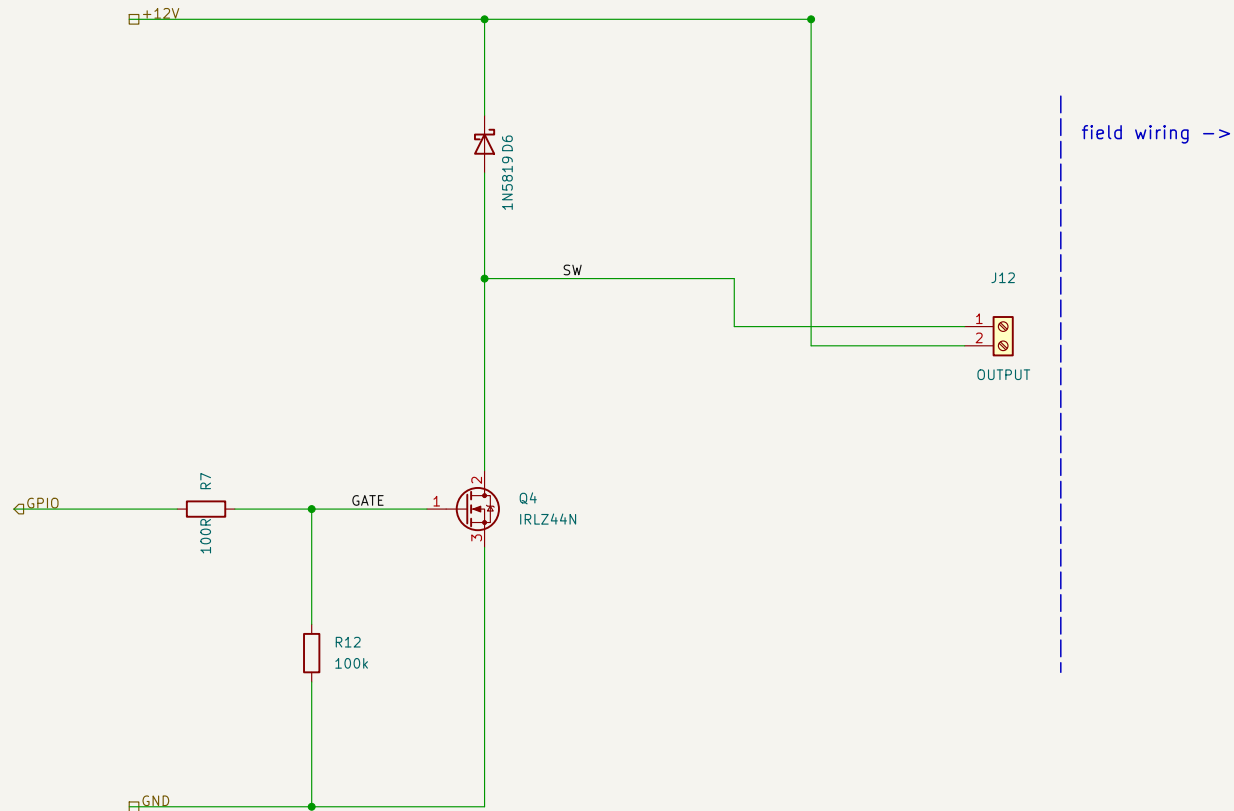
Date:

Rev:

Id: 4/6

ONE PUMP / LOAD CHANNEL

Drawn once here and used twice times – see the root sheet.



GPIO, +12 V and GND are the only connections that leave this sheet. Everything else is internal, which is why the root sheet can draw the whole channel as one block.

RPD (100k, gate to GND) is a SAFETY part, not decoration: ESP32 GPIOs float before setup() runs and through a reset or a brownout, and this resistor is what holds the output off across that window.

Active HIGH is fixed by the firmware – Actuator_Digital::act() is a bare digitalWrite(pin, value) with no invert flag. A low-side N-MOSFET is the natural match; most cheap opto-relay boards are active LOW and would invert every output.

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Sheet: /Pump/
File: driver_switch.kicad_sch

Title:

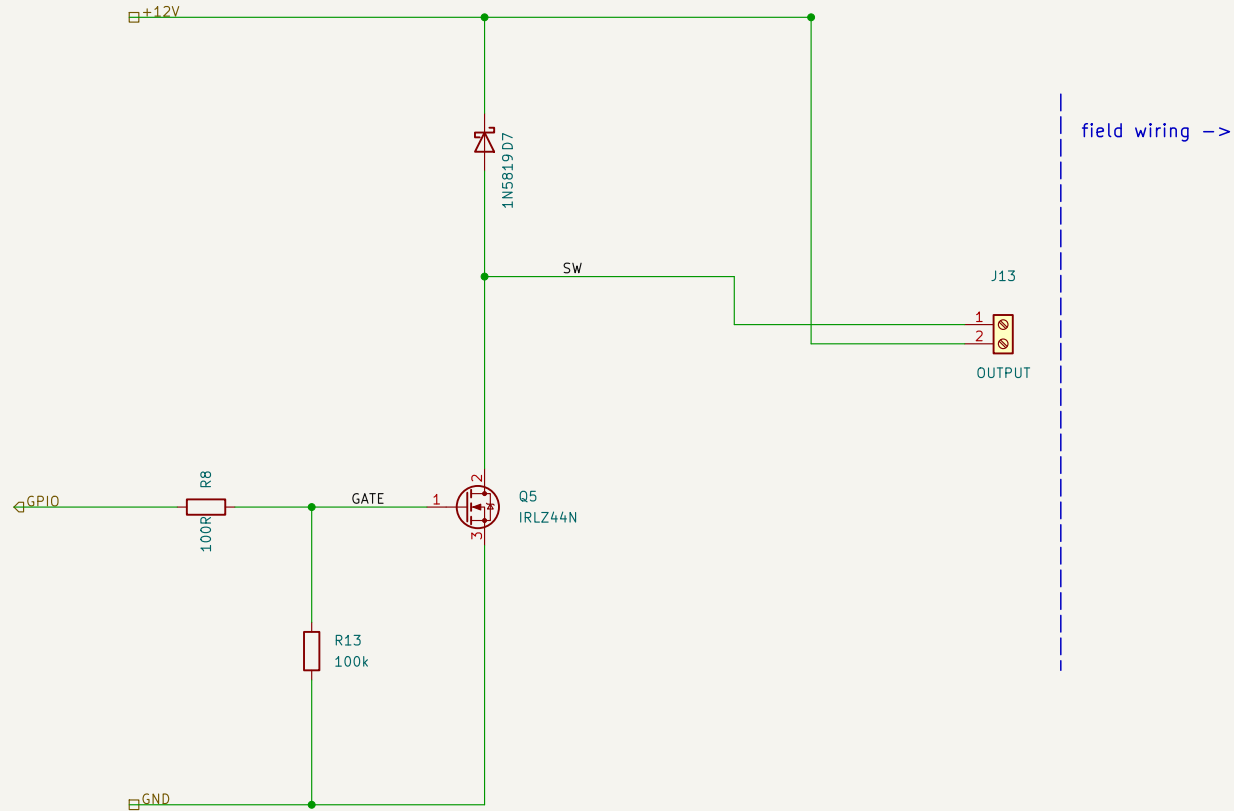
Size: A4
KiCad E.D.A. 10.0.4

Date:

Rev:
Id: 5/6

ONE PUMP / LOAD CHANNEL

Drawn once here and used twice times – see the root sheet.



GPIO, +12 V and GND are the only connections that leave this sheet. Everything else is internal, which is why the root sheet can draw the whole channel as one block.

RPD (100k, gate to GND) is a SAFETY part, not decoration: ESP32 GPIOs float before setup() runs and through a reset or a brownout, and this resistor is what holds the output off across that window.

Active HIGH is fixed by the firmware – Actuator_Digital::act() is a bare digitalWrite(pin, value) with no invert flag. A low-side N-MOSFET is the natural match; most cheap opto-relay boards are active LOW and would invert every output.

D clamps the inductive kick when the FET turns off. Cathode to +12 V – backwards, it is a short across the supply the moment the output goes on.

Sheet: /Load/
File: driver_switch.kicad_sch

Title:

Size: A4
KiCad E.D.A. 10.0.4

Date:

Rev:
Id: 6/6